

1 Sketches of the System

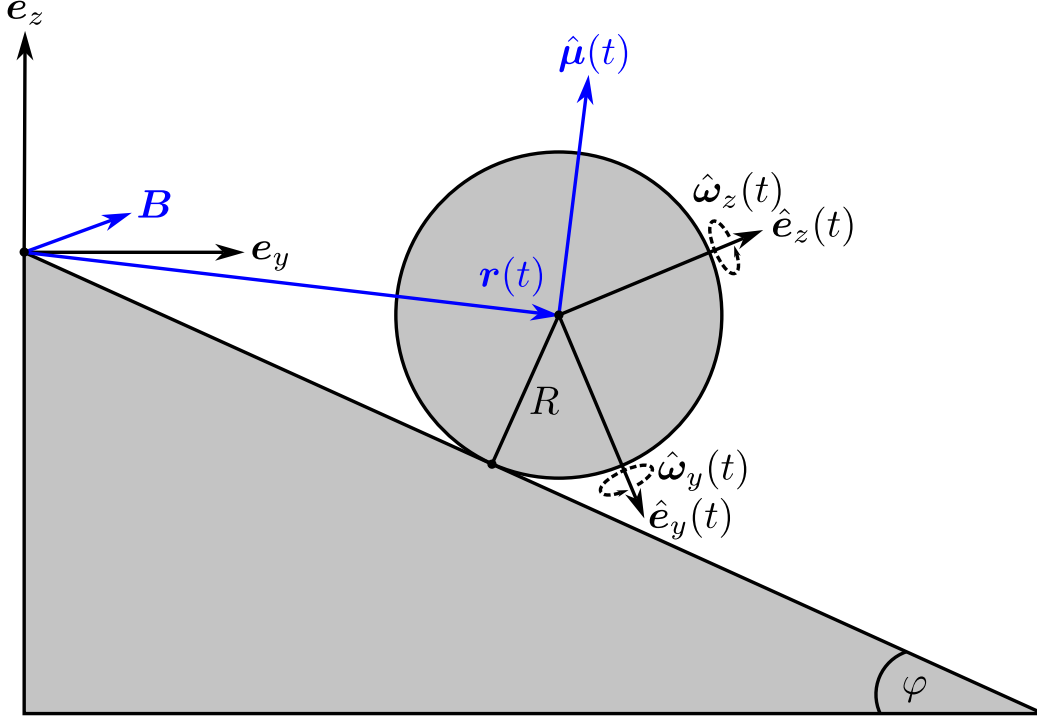


Figure 1: Depicted is the yz -projection of the incline system, the z -direction is orthogonal to the paper plane. The incline has an inclination angle φ and the sphere is of radius R . Two orthonormal coordinate frames are considered: The fixed frame $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$ and the body frame $\{\hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z\}$. Vectors expressed in the body frame carry hats and are time-dependent, stationary vectors not. Blue vectors might have a non-zero z -component and are therefore only projections. The outer magnetic field is represented by $\mathbf{B}$, the magnetic dipole vector of the sphere by $\hat{\boldsymbol{\mu}}(t)$, and the centre of mass measured from the fixed frame by $\mathbf{r}(t)$. The in-plane angular velocities of the body frame are shown as $\hat{\omega}_x$ and $\hat{\omega}_y$.

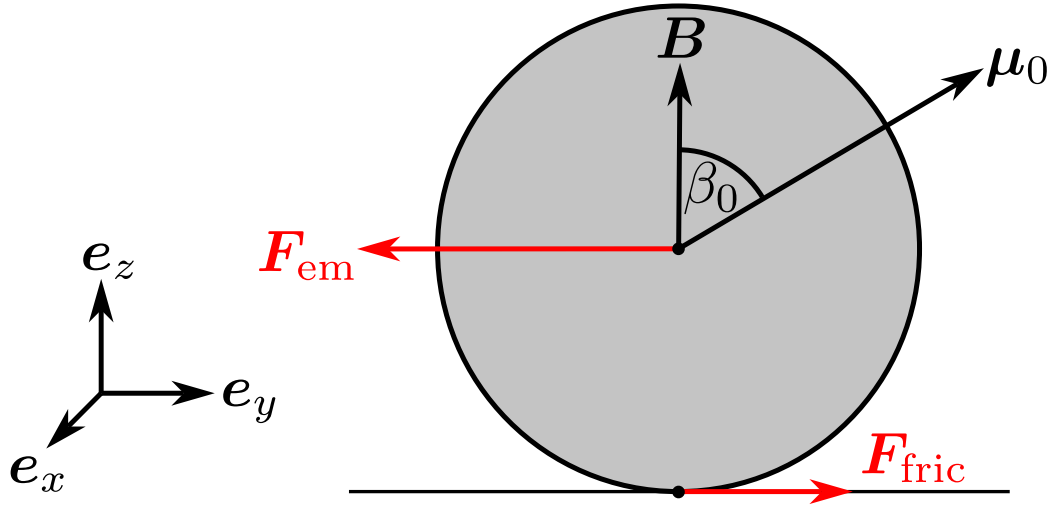


Figure 2: Shown is a sketch of the flat system with sufficiently small initial displacement angle β_0 between the outer magnetic field $\mathbf{B}$ and the initial magnetic dipole vector $\boldsymbol{\mu}_0$. Note that the system is effectively reduced to two dimensions. The red vectors indicate the oscillation-driving electromagnetic force $\mathbf{F}_{\text{em}}$ and the damping frictional force $\mathbf{F}_{\text{fric}}$. The frictional force includes aerodynamical drag and rolling resistance.

2 Figure Family A

These figures are results of a proof-of-concept toy model: The incline without electromagnetic interaction. The goal is to compare numerical results to analytical equations.

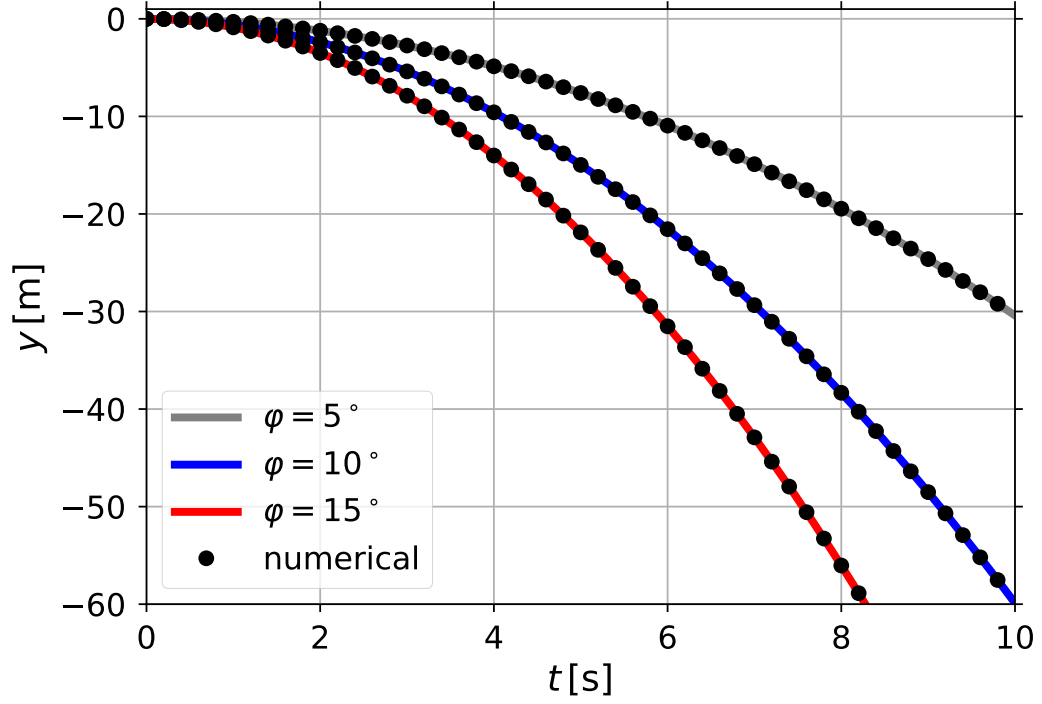


Figure A1: Shown are different evolutions of the y -projection of the centre-of-mass coordinate $\mathbf{r} = (x, y, z)^\top$ over time t for different incline angles φ without friction. Lines were calculated from the analytical solution and compared to the numerical results, represented as dots.

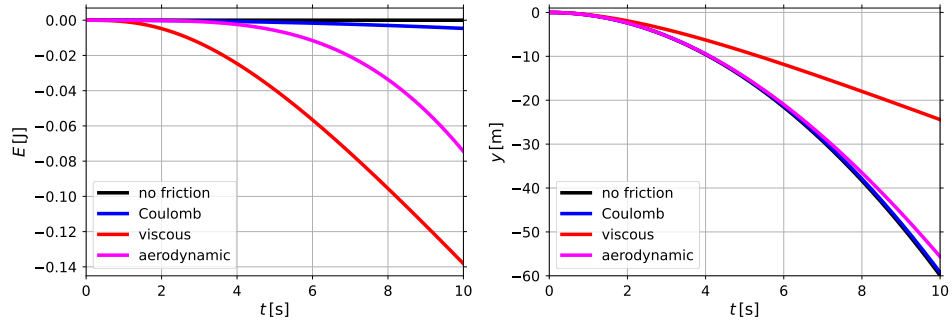


Figure A2/A3: The left plot shows the energy evolution $E(t)$ of the system over time t , while the right one depicts the y -COM-coordinate evolution. The data originates from the numerical simulation program. Compared are different friction models, where *Coulomb* represents dry rolling resistance, *viscous* represents a linear damping approximation to rolling resistance, and *aerodynamic* represents spherical aerodynamic drag.

3 Figure Family B

4 Figure Family C

5 Figure Family D

6 Figure Family E